

DEVELOPING GENE THERAPY FOR CONTROLLING CANCER

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Introduction to Cancer

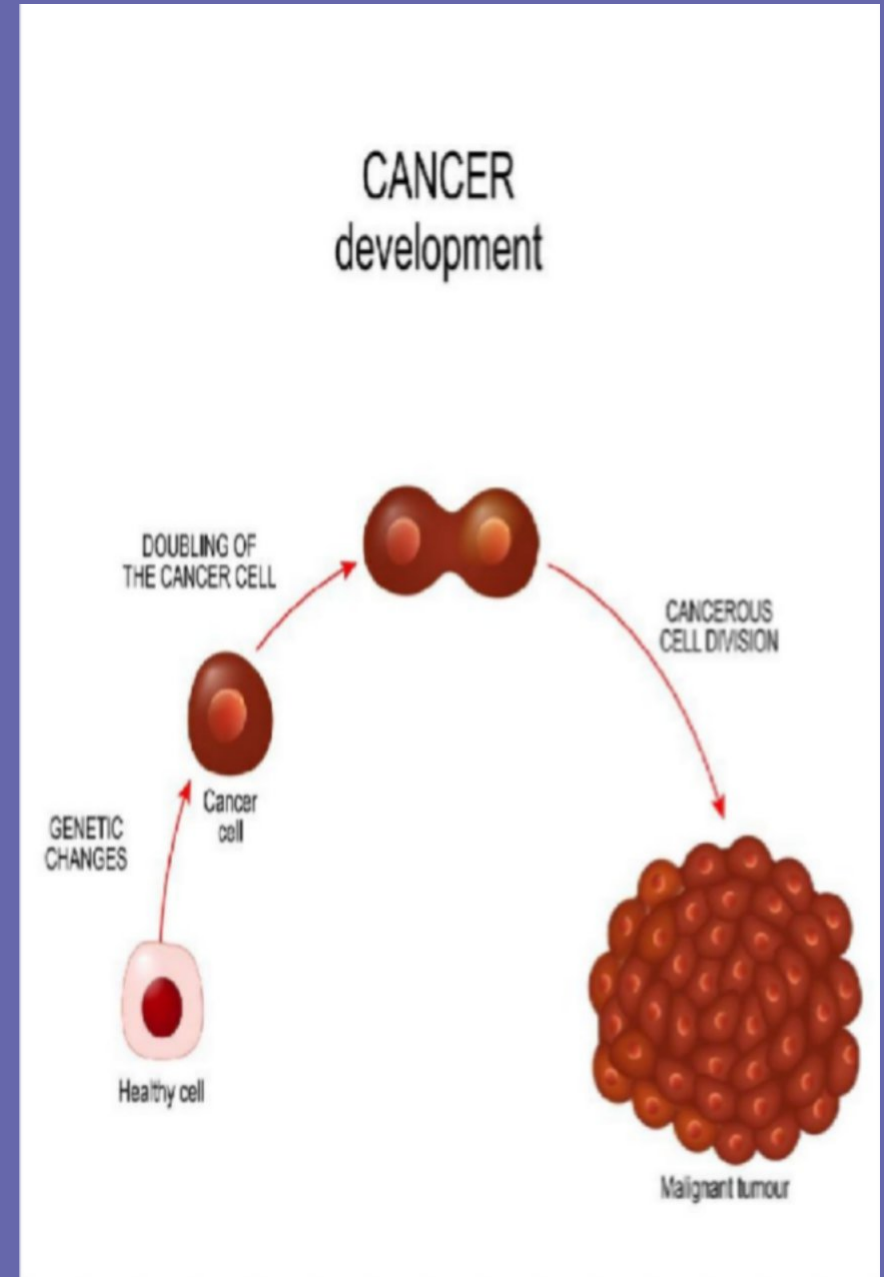
- Cancer is a disease caused by uncontrolled cell growth.
- It can spread to different body parts.

Common types:

- Breast cancer
- Lung cancer
- Blood cancer

Traditional treatments :-

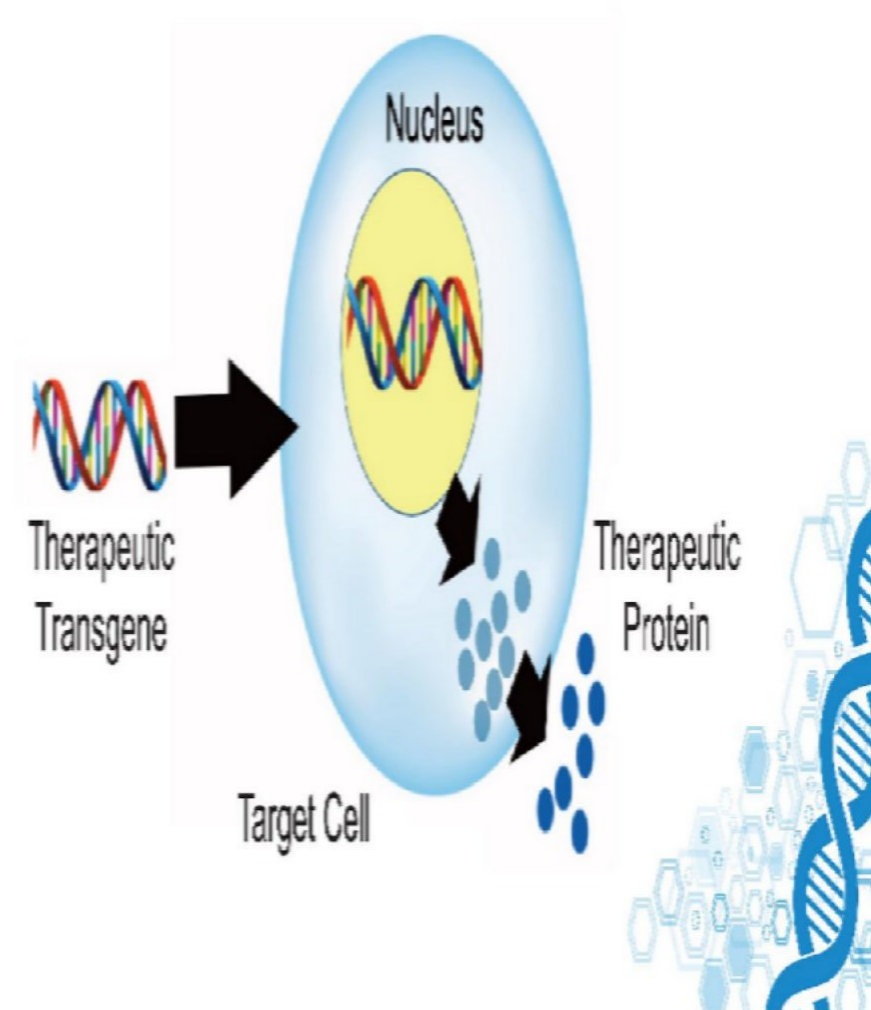
- Surgery
- Chemotherapy
- Radiation therapy



What is Gene Therapy?

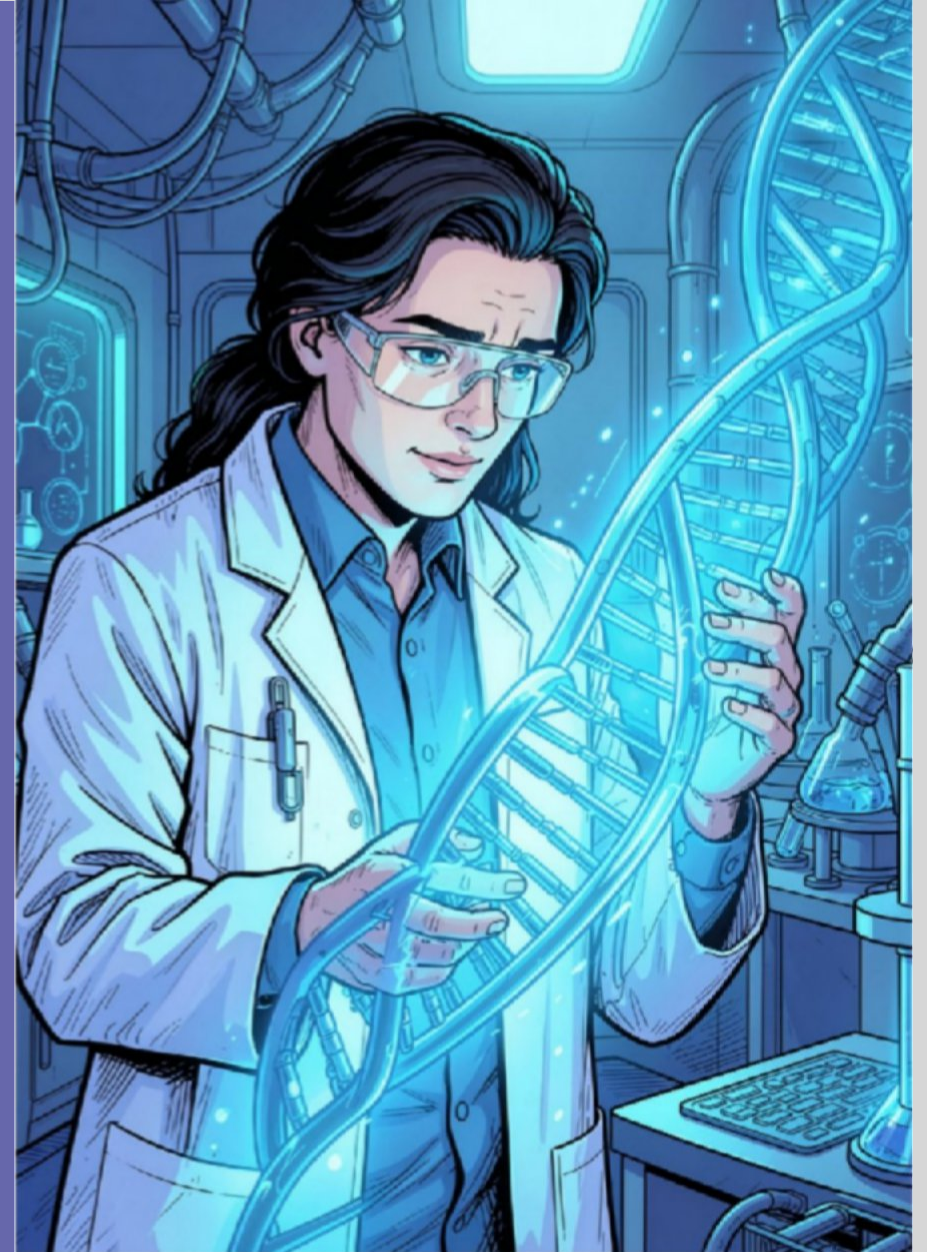
- Gene therapy is a modern medical technique.
- It modifies genes to treat diseases.
- Healthy genes are inserted into cells to fight cancer.
- It targets the root cause of the disease.

The Principle of Gene Therapy



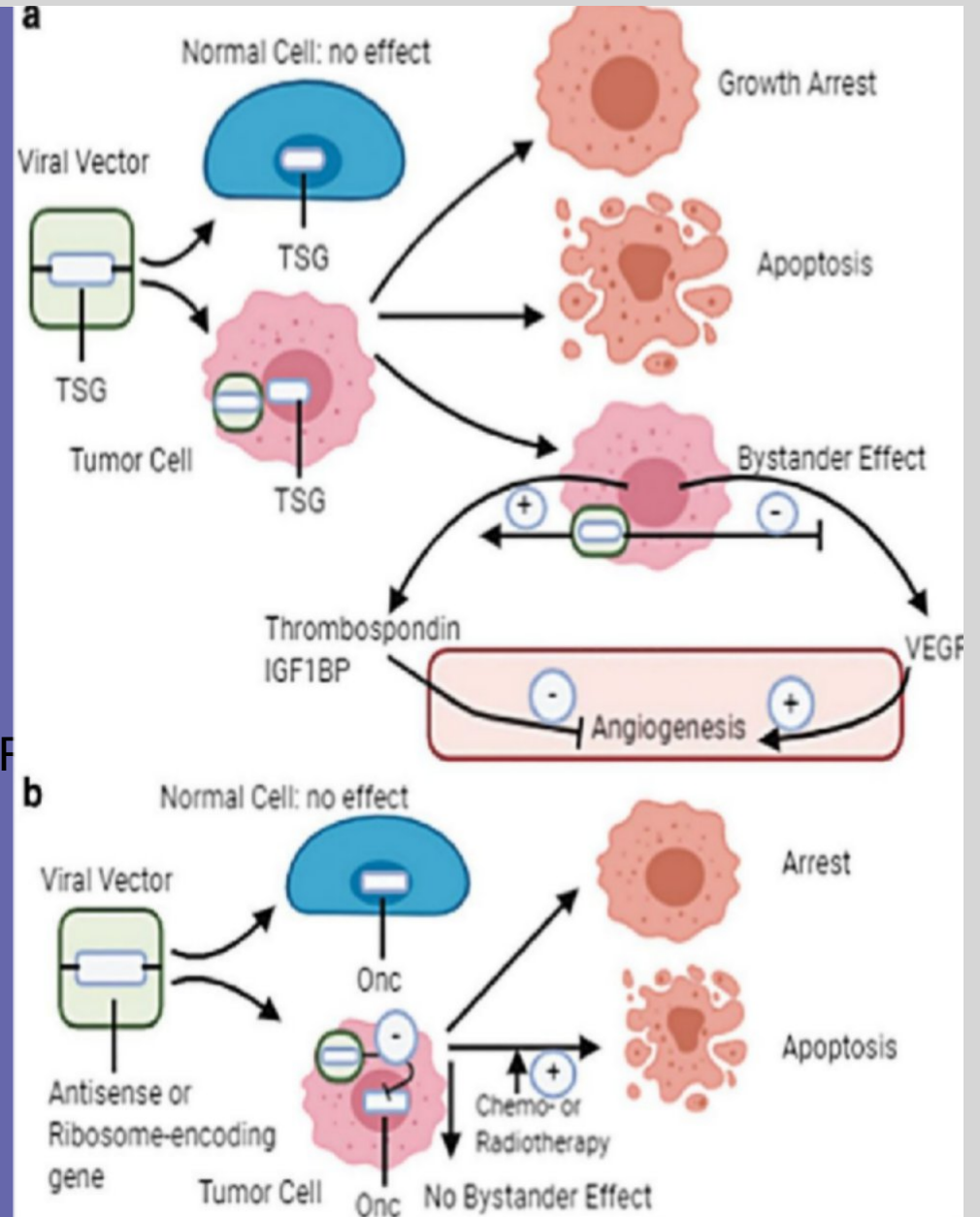
Objectives of Gene Therapy in Cancer

- Destroy cancer cells
- Stop tumor growth Improve immune response
- Reduce side effects of chemotherapy
- Provide personalized treatment
 - stop tumor growth
 - improve immune response
 - reduce side effects of chemotherapy
- provide personalized treatment



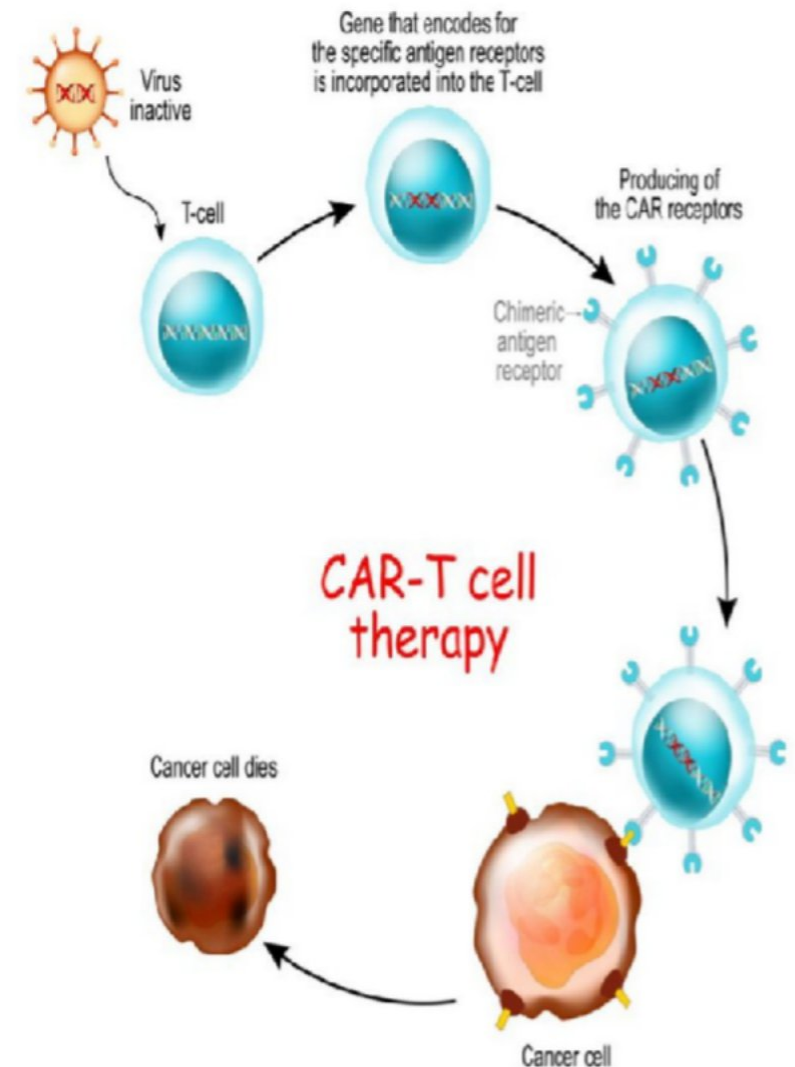
HOW GENE THERAPY WORKS

- IDENTIFY DEFECTIVE GENES
- INSERT THERAPEUTIC GENES
- USE VECTORS (VIRUSES) TO DELIVER GENES
- MODIFIED CELLS ATTACK CANCER CELLS



TYPES OF GENE THERAPY

- **Somatic Gene Therapy**
 - Targets body cells
 - Effects are not inherited
- **Germline Gene Therapy**
 - Targets reproductive cells
 - Effects can pass to future generations
- **Technologies Used**
 - CRISPR-Cas9
 - Viral vectors
 - Nanotechnology



Advantages of Gene Therapy

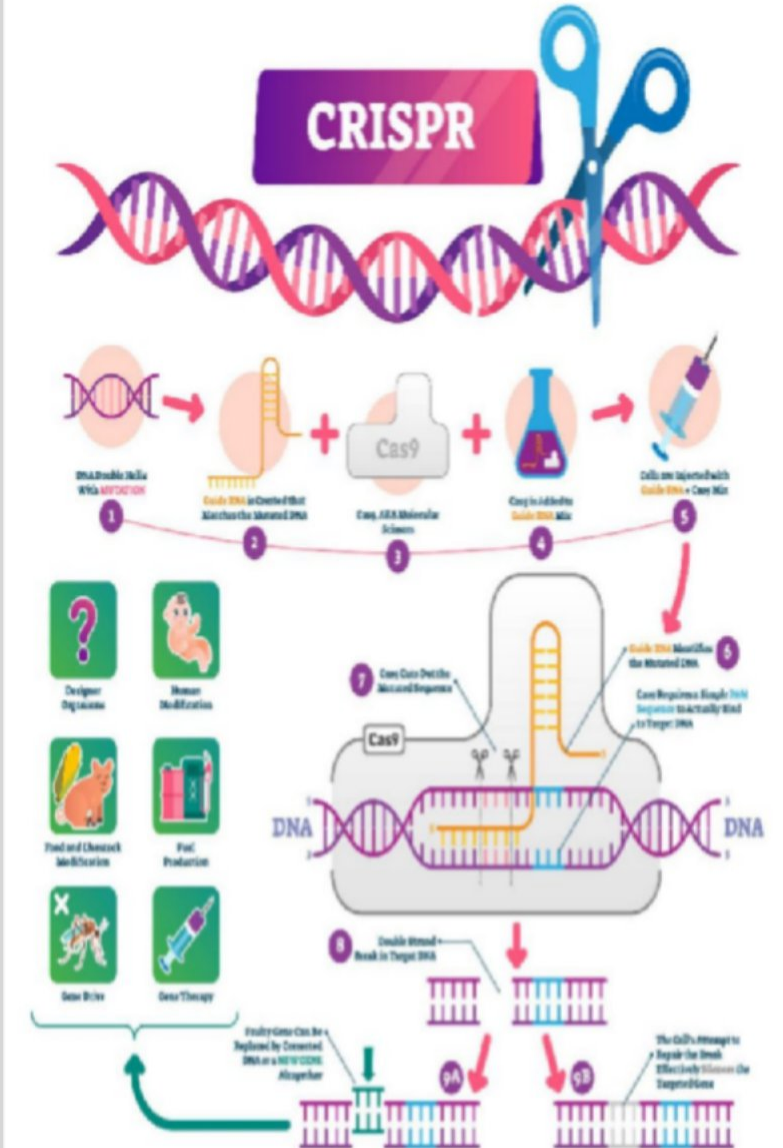
- Precise treatment
- Fewer side effects
- Targets cancer cells directly
- Potential long-term cure
- Personalized medicine

❑ CHALLENGES

- HIGH COST
- ETHICAL ISSUES
- IMMUNE REACTIONS

❑ RECENT DEVELOPMENTS

- CAR-T CELL THERAPY
- SUCCESSFUL BLOOD CANCER TREATMENT
- ADVANCED CLINICAL TRIALS



Thank you